MILITARY SPECIFICATION

DRIVE, ANTENNA, PATTERN CONTROL, FOR AS-473/ASB-1 (BuOrd Drawings 776376, 776520)

1. SCOPE

1.1 This specification covers the equipment, material, and performance requirements to be used in the manufacture of an Antenna Pattern Control Drive for use in a radar scanning antenna. One type and one grade only is specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification.

2.1.1 Specifications

Military

MIL-P-116 Preservation, Methods of

2.1.2 Standards

Military

MIL-STD-129 Military Standard Marking of Shipments

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2.1.3 Drawings

Bureau of Ordnance, Department of the Navy

List of Drawings, LD 272215, Pattern Control Drive Unit and all drawings and specifications listed thereon.

2.1.4 Publications

Bureau of Ordnance, Department of the Navy

NAVORD OP 400 — Manufacture and Inspection of Ordnance Material, General Specification for

NAVORD OD 8176 — Paragraphs applicable to this equipment

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the Procuring Agency or as directed by the Contracting Officer)

2.1.5 Other Publications

National Bureau of Standards Publications

Handbook H28 — Screw-Thread Standards for Federal Services

(Copies of H28 may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Type Tests - Type tests as specified in Section 4 of this specification shall be made on representative samples of production assemblies.

3.2 General - Equipment furnished under this specification shall meet applicable requirements listed in NAVORD OD 8176. In cases of conflict, this specification shall apply.

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3.3 Material - Materials shall conform to applicable specifications as specified herein and on the drawings. When materials are used which are not specifically designated, they shall be of the best commercial quality and entirely suitable for the purpose.

3.4 Design and Construction - The assembly shall conform to and be approved in accordance with this specification and applicable specifications, standards, drawings, and publications.

3.5 Interchangeability - Interchangeability shall be insured by conformance with dimensional limitations set forth on outline drawing, BuOrd number 776761.

3.6 Nameplates - A nameplate, permanently and legibly marked and made in accordance with BuOrd drawing 776416, shall be attached to each assembly.

3.7 Performance

3.7.1 Torque - Torque required to slip the clutch shall be 21 ± 4 inch-ounces.

3.7.2 The torque required to rotate the lead screw shall be 3 ± 1 inch-ounces when no power is supplied to the motor.

3.8 Environmental Conditions

3.8.1 Temperature - The equipment shall conform with all requirements of this specification under temperature conditions as specified in NAVORD OD 8176 except that the temperature range shall be modified to read -55°C to +71°C (-67°F to +160°F).

3.8.1.1 To determine compliance with 3.8.1, each type test sample shall be subjected to inspection tests 4.3.3.2.2 and 4.3.3.2.3 at maximum and minimum temperatures.

3.8.2 Humidity - The equipment shall conform with all requirements of this specification under humidity conditions as specified in OD 8176.

3.8.2.1 To determine compliance with 3.8.2, each type test sample shall be subjected to inspection tests 4.3.3.2.2 and 4.3.3.2.3 at combinations of temperature and humidity that will alternately form water and frost on the unit.

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3.8.3 Vibration - The equipment shall be subjected to vibration conditions as specified in OD 8176 except that the total excursion shall not exceed 0.031 inch. The bracket, BuOrd drawing 837282, carried by the follower nut, BuOrd drawing 837287, shall be secured to prevent damage to the unit. See BuOrd drawing 776761 for normal mounting.

3.8.4 Shock - The equipment shall not suffer damage or subsequently fail to provide the performance specified herein when it is mounted as normally used and is subjected to six (6) impact shocks of 10g's, each shock impulse having a time duration of 11 ± 1 milliseconds. See BuOrd drawing 776761 for normal mounting. Two (2) shocks shall be applied in each of the three (3) principal axes, one (1) shock in each direction.

3.9 Lubrication - Lubrication shall be as specified on drawings.

3.10 Alignments - Alignments shall be made in accordance with procedure specified in 4.3.3.2.1.

3.11 Workmanship - Workmanship shall be in accordance with the best instrument practice. Particular attention shall be given to machined surfaces to insure conformance with drawing limitations and against superficial defects. Attention shall also be given to painted areas to insure clean and scratch-free surfaces.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 General Instructions

4.1.1 Responsibility for Tests - The Contractor shall furnish all samples, if required, and shall be responsible for accomplishing the inspections and tests specified herein.

4.1.2 Previous Approval - Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.1.3 Supervision of Tests - When inspection is conducted at the Contractor's plant, all inspection and testing shall be under the supervision of a Government Inspector.

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4.1.4 Record of Tests - The Contractor shall maintain a record available to the Inspector, showing the results of all Inspection Tests.

4.1.5 Laboratory Facilities - Contractors not having laboratory facilities satisfactory to the Inspector shall engage the services of a commercial testing laboratory acceptable to the Inspector.

4.1.6 Pre-Submission Testing - No item, part, or complete equipment shall be submitted to the Government Inspector by the Contractor until it has been previously tested and inspected by the Contractor and found to comply to the best of his knowledge and belief with all applicable requirements. With the consent of the Contractor and at the discretion of the Procuring Agency, this prior test and inspection may be participated in or witnessed by the Government Inspector with the object of eliminating the necessity of repeating such test and inspection after the equipment has been formally submitted to the Government Inspector.

4.1.7 Rejection and Retests - The equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Agency.

4.2 Inspection

4.2.1 Examination of Product - Each assembly, subassembly, and part shall be examined as the Inspector may deem necessary to determine conformance with this specification and the applicable drawings with respect to material, dimensions, instructions, and workmanship.

4.3 Test Procedures

4.3.1 Classification - Tests shall be classified as follows:

a. Type Tests
b. Inspection Tests

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4.3.2 Type Tests - Type tests shall be performed on one (1) of the first twenty (20) production units and on one (1) of each subsequent group of one hundred (100) production units or parts thereof, for each manufacturer supplying equipment in accordance with this specification.

4.3.2.1 Type test samples shall be selected at random from production units that have passed all inspection tests and shall have inspection tests 4.3.3.2.2 and 4.3.3.2.3 repeated under environmental conditions specified in 3.8 of this specification.

4.3.3 Inspection Tests - Inspection tests shall be made on all production assemblies submitted for acceptance under the Contract, to determine conformance with the requirements of this specification and to determine that the assemblies are equivalent in performance and construction to the approved samples. All production lots as defined in 4.3.2 will be accepted contingent upon type tests approval.

4.3.3.1 Test Equipment - The following listed equipment or their equivalents will be required to perform tests specified in 4.3.3.2.

4.3.3.1.1 Source of power, 115 volts, 400 cycles, single phase.

4.3.3.1.2 Test Stand. See Figure 1.

4.3.3.1.3 Spring scale.

4.3.3.1.4 Voltmeter.

4.3.3.2 Tests - All assemblies shall be subjected to tests specified in 4.3.3.2.1 through 4.3.3.2.3 at normal room conditions.

4.3.3.2.1 Alignments

4.3.3.2.1.1 Check potentiometer and determine that the contacting arm does not reach the limit stops before the stop lugs of the lead screw engage with the stop lugs of the follower nut.

4.3.3.2.1.2 Check perpendicular distance from the center of the dowel pin to the centerline of the lead screw. This must conform with dimension as shown in Figure 2.

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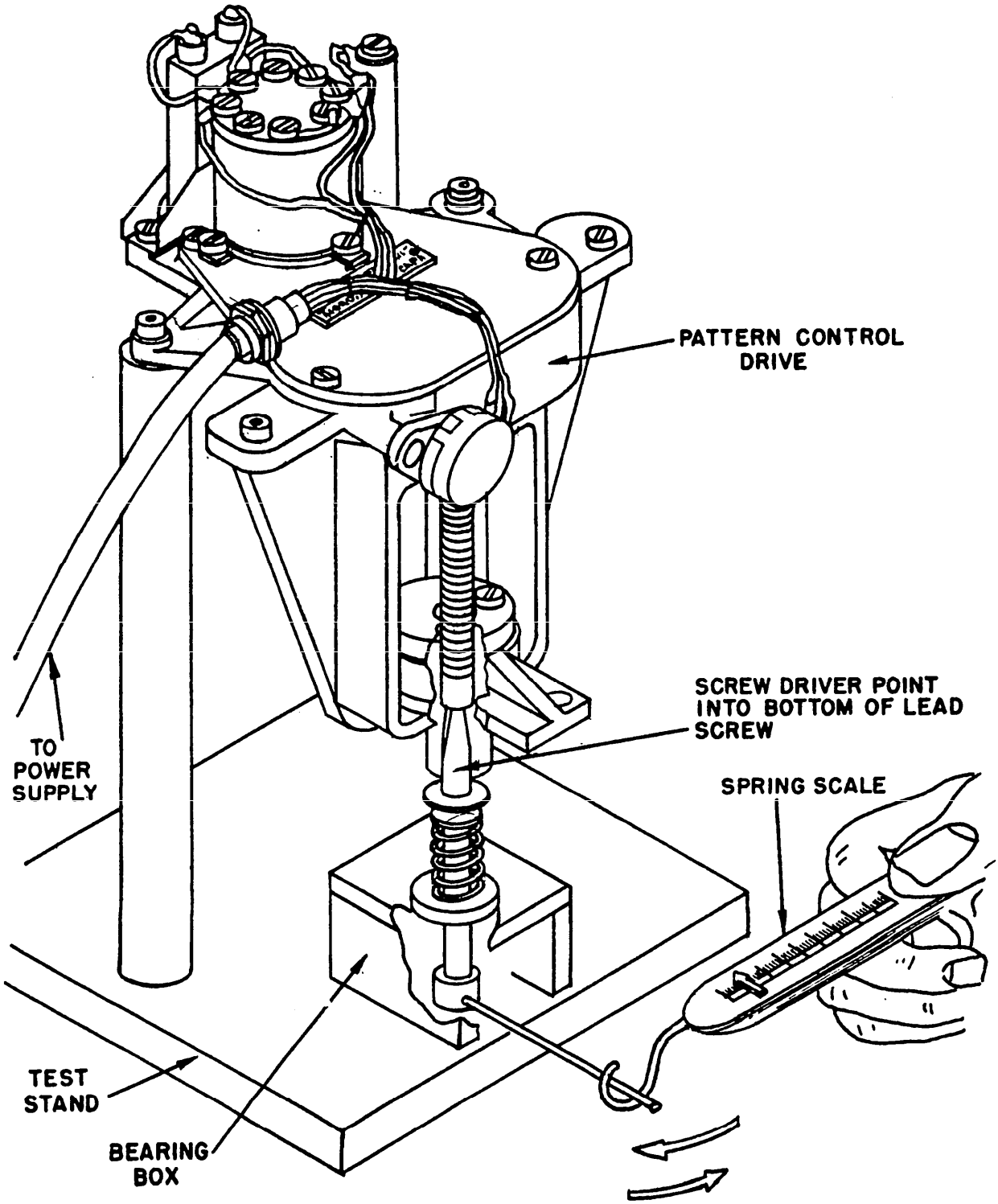


Figure 1. Pictorial Schematic of Pattern Control Drive Test

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4.3.3.2.1.3 Check lead screw and follower nut stops at each end of travel and determine that overlap is in accordance with drawing requirements. The overlap must provide a positive stop.

4.3.3.2.1.4 Check mounting dimensions "A," "B," and "C" as shown in Figure 2.

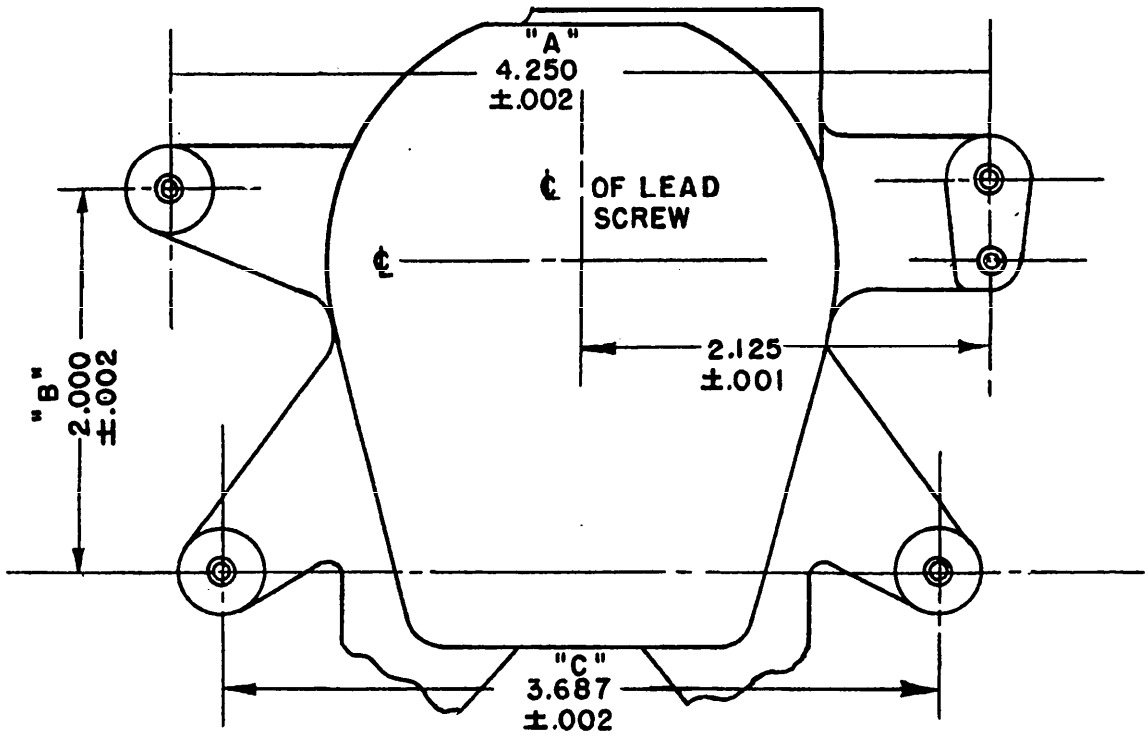


Figure 2. Mounting Dimensions

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4.3.3.2.1.5 Check the lead screw for vertical alignment as shown in Figure 3, Item "G."

4.3.3.2.1.6 Check axial play between the link and nut as shown in Figure 3, Item "F."

4.3.3.2.1.7 Check the mounting hole dimensions of the link, Items "D" and "E" of Figure 3.

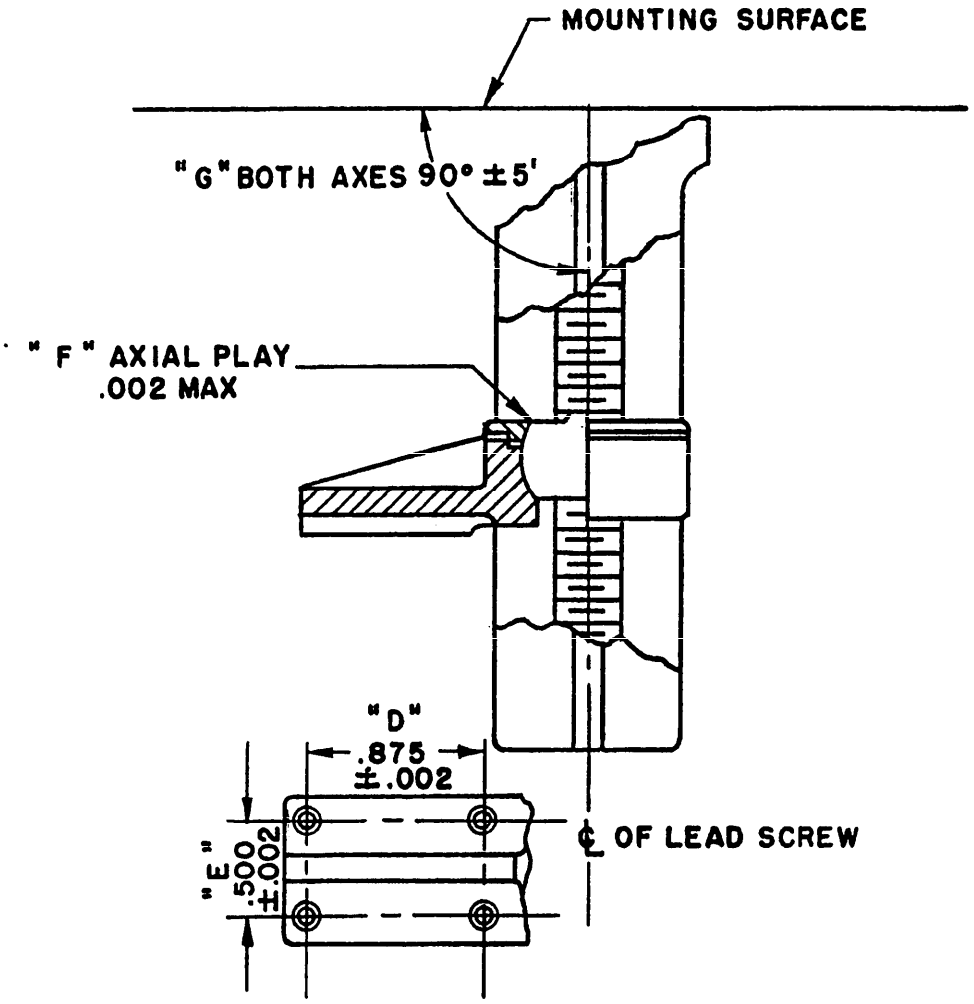


Figure 3. Reference Points

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CAUTION: Manually turn lead screw to check limit stops to prevent possible damage to potentiometers due to improper setting.

4.3.3.2.2 Directional Sense - Check the assembly for proper directional sense through the use of the test set-up as illustrated by Figure 4. The plugs are connected and voltage is applied to the system as shown. The toggle switch is thrown to the "RAISE" position and the direction of travel of the follower nut is noted. The nut shall travel to its uppermost position, the position nearest the potentiometer. When the nut is in its uppermost position, the voltmeter shall read five (5) or less. The switch is then thrown to the "LOWER" position. The follower nut shall move to opposite end of lead screw. When follower nut is in its lowest position, the voltmeter shall read 110 volts or more.

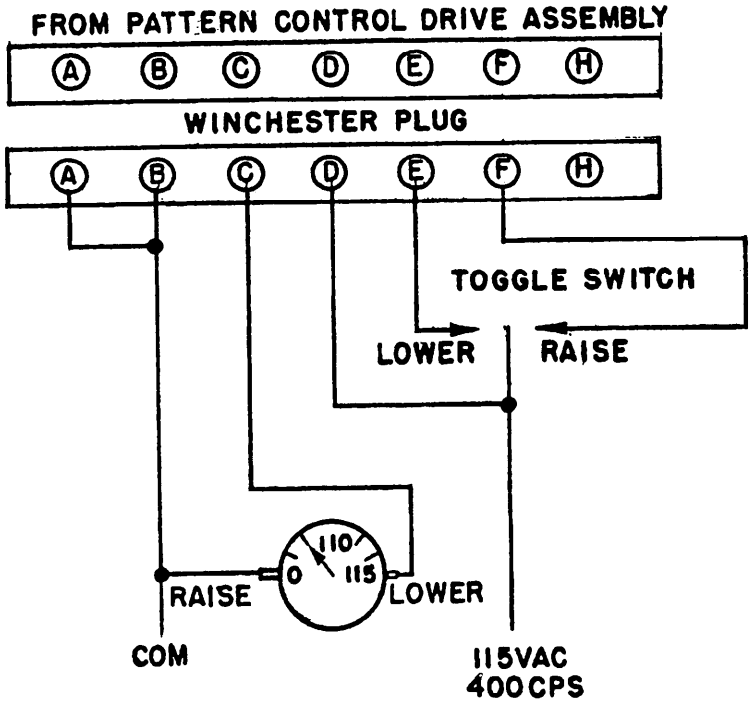


Figure 4. Wiring Diagram for Test Setup

4.3.3.2.3 Torque Test - Check the torque required to turn the lead screw with no power applied to the motor. This torque shall be as specified in 3.7.2. Subtract this torque from 25 inch-ounces. Apply power (115 volts, 400 cycles) to motor. The output torque then shall be equal to the difference of the above subtraction +0.0, -8.0 inch-ounces.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing - Packaging and packing shall be in accordance with MIL-P-116 , Method II, paragraph 3.5.7.2.2 or 3.5.7.2.4, as applicable. Such items shall be packaged by either of the above methods as directed by the Procuring Agency. Further, metal containers shall be surrounded by suitable cushioning material and packed in a wooden case as directed by the Procuring Agency.

5.1.1 Labeling and Marking - Labeling and marking shall be in accordance with MIL-STD-129, Military Standard Marking of Shipments, except as may be modified by the Procuring Agency.

6. NOTES

6.1 Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawing, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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